

No-clustering bootstrap screen $\text{Sharpe} > 0.5$ (IS 2012–2022)

Directed factor-mimicking portfolio catalogue

Abstract

This catalogue constructs one FMP for each configured field or composite and compares all factors on one factor- and control-complete monthly window. Factors are indexed by reported raw Sharpe, highest first. Direction is frozen upstream; this report performs no return-based sign selection.

Methodology

Signals are observed at month-end, earn the next calendar month's simple return, and refresh monthly. Every factor must independently pass breadth, history, coverage, and non-degeneracy gates; none is silently dropped. One factor- and control-complete window fixes the reported Sharpe order and feeds all reported analyses. Displayed signal signs define the pre-direction composite; the reported Direction then orients the whole FMP. Risk Loadings include HAC t-statistics; Return Contributions are loading times annualized control mean, plus alpha and realized total. Rank diagnostics use 0/1/3/6/12/36-month horizons; available-history statistics are diagnostic only. Return Uncertainty uses paired circular stationary-bootstrap draws of the common monthly panel. It measures sampling and path uncertainty, not out-of-sample generalisation.

Construction Configuration

Setting	Effective value
Factors	21
Universe	shared security screen; refreshed monthly
Formation filters	price $\geq$ None; 21d dollar volume $\geq$ none
Market-cap screen	top 90.0% of names monthly
Signal preparation	within-month winsorize, normalize, then demean
Composite	mean available normalized components; factor-specific coverage floor
Construction	one blanket profile
Rebalance / return	month-end / next calendar month simple return
Costs	effective configured costs; default 0 bps
Minimum sample	20 stocks/month; 24 common months; 90% factor coverage
Comparison window	2012-02-01 to 2022-12-01; factor- and control-complete
Winsor settings	1%-99%
Controls	MKT, SMB, VAL, MOM, BAB, QMJ
Return uncertainty	paired stationary bootstrap; 2000 draws; expected block 12 months; seed 0
Catalogue note	Each of the 21 admitted signals is one standalone factor. No clustering, PCA, varimax, or post-screen factor pruning is applied. Admission requires at least 70% of fixed-orientation 2012–2022 stationary-bootstrap Sharpes to exceed 0.5.

Returns PCA

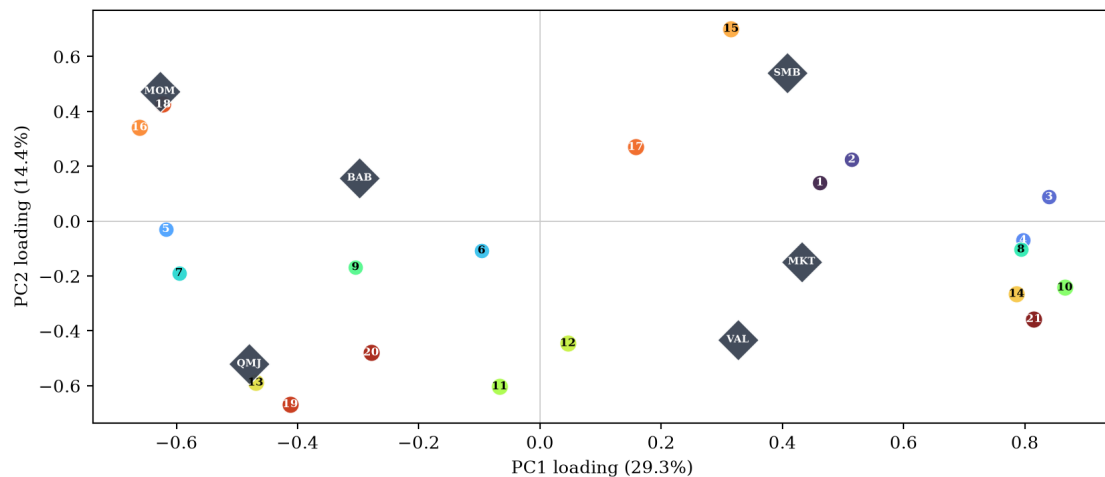


Figure: catalogue factors are numbered by reported-Sharpe index; attribution controls are named; $N=131$ common months.

Index

Idx	Factor	Dir.	N
1	ret_zscore_63d	+1	131
2	resid_ret_252v_1260c_SPY	+1	131
3	stoch_k_14d	+1	131
4	bb_pctb_20d	+1	131
5	eps_growth_yoy	+1	131
6	capex_to_revenue	+1	131
7	net_income_growth_yoy	+1	131
8	rsi_14d	+1	131
9	dpo	+1	131
10	price_zscore_63d	+1	131
11	debt_to_equity	+1	131
12	debt_to_assets	+1	131
13	gp_to_equity	+1	131
14	momentum_0_21	+1	131
15	accruals_total	+1	131
16	momentum_21_252_vol_63d	+1	131
17	accruals_noa_proxy	+1	131
18	momentum_21_252	+1	131
19	share_turnover_21d	+1	131
20	asset_turnover	+1	131
21	ma_dist_63d	+1	131

Returns

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ret_zscore_63d	8.24%	5.21%	1.58	4.51	-4.40%	0.179	7.32%	5.21%
resid_ret_252v_1260c_SPY	8.40%	5.88%	1.43	3.53	-4.59%	0.197	8.37%	5.88%
stoch_k_14d	8.96%	6.90%	1.30	4.22	-8.91%	0.294	10.04%	6.90%
bb_pctb_20d	8.13%	6.49%	1.25	4.20	-7.53%	0.189	9.00%	6.49%
eps_growth_yoy	5.49%	5.14%	1.07	3.04	-11.71%	0.241	2.72%	5.14%
capex_to_revenue	4.42%	4.53%	0.98	3.38	-5.61%	0.263	2.40%	4.53%
net_income_growth_yoy	4.55%	4.67%	0.97	2.88	-11.44%	0.309	1.63%	4.67%
rsi_14d	5.72%	6.20%	0.92	3.26	-12.74%	0.220	7.76%	6.20%
dpo	4.05%	4.43%	0.92	3.03	-8.94%	0.158	2.02%	4.43%
price_zscore_63d	5.57%	6.65%	0.84	2.83	-7.14%	0.340	8.57%	6.65%
debt_to_equity	4.36%	5.28%	0.83	2.62	-8.76%	0.445	2.77%	5.28%
debt_to_assets	4.05%	4.97%	0.81	2.68	-7.38%	0.252	3.12%	4.97%
gp_to_equity	4.40%	5.41%	0.81	2.51	-8.67%	0.497	0.44%	5.41%
momentum_0_21	5.26%	6.47%	0.81	3.05	-9.68%	0.309	7.40%	6.47%
accruals_total	3.92%	5.23%	0.75	2.62	-7.17%	0.394	5.70%	5.23%
momentum_21_252_vol_63d	5.52%	7.95%	0.69	2.81	-10.16%	0.679	0.90%	7.95%
accruals_noa_proxy	3.38%	4.99%	0.68	2.19	-7.68%	0.241	4.32%	4.99%
momentum_21_252	5.60%	8.38%	0.67	2.87	-10.23%	0.678	1.09%	8.38%
share_turnover_21d	5.54%	8.54%	0.65	2.05	-31.05%	0.463	4.89%	8.54%
asset_turnover	4.05%	6.26%	0.65	2.58	-7.85%	0.487	-0.82%	6.26%
ma_dist_63d	4.00%	7.57%	0.53	1.94	-11.14%	0.438	7.66%	7.57%

Return Uncertainty

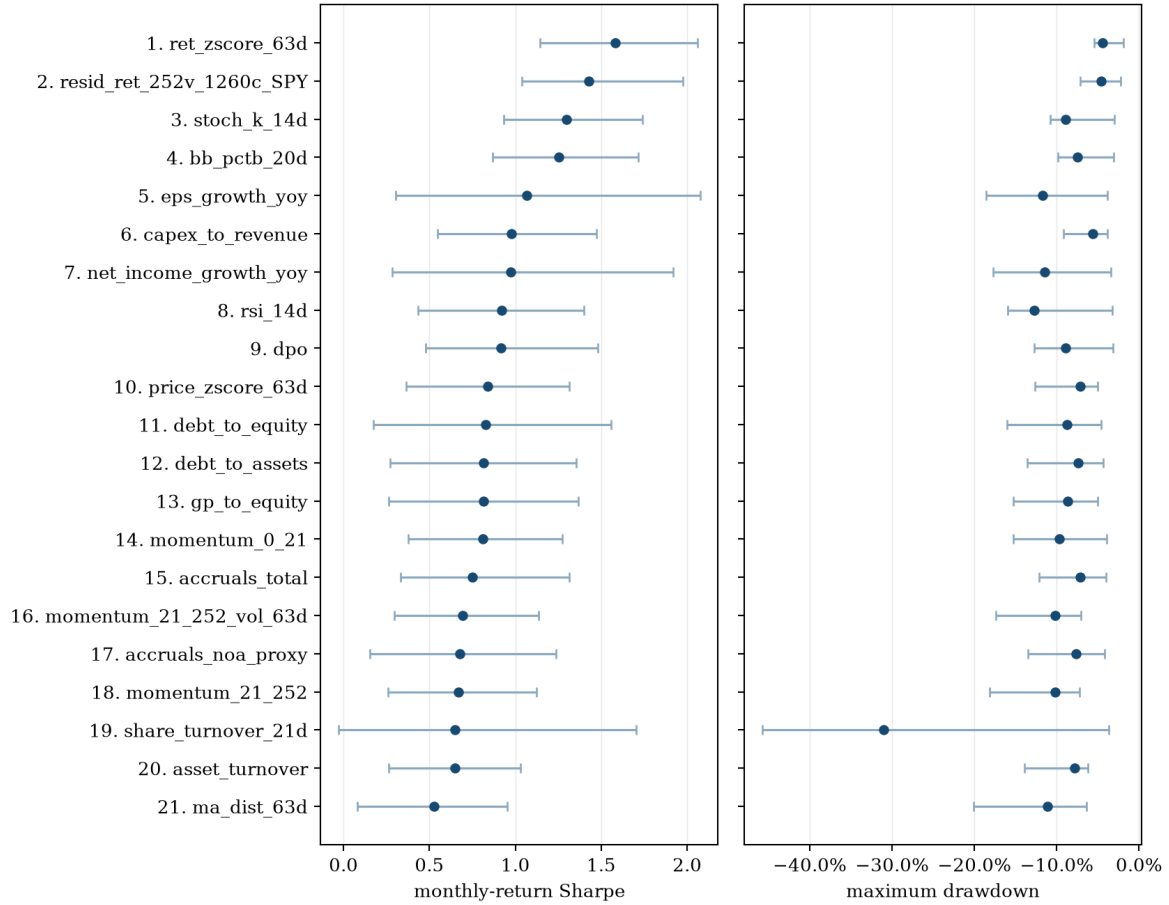


Figure: point estimates and paired stationary-bootstrap 95% intervals.

Idx	Factor	Ann. ret. [95%]	Sharpe [95%]	Pr(S<=0)	Skew [95%]	Max DD [95%]	ES 5% [95%]
1	ret_zscore_63d	8.2% [5.0%, 11.8%]	1.58 [1.14, 2.06]	0.0%	2.29 [0.50, 3.05]	-4.4% [-5.5%, -1.9%]	-1.5% [-1.9%, -1.1%]
2	resid_ret_252v_1260c_SPY	8.4% [4.5%, 13.2%]	1.43 [1.04, 1.97]	0.0%	2.96 [-0.03, 3.51]	-4.6% [-7.1%, -2.3%]	-1.8% [-2.1%, -1.3%]
3	stoch_k_14d	9.0% [5.4%, 13.2%]	1.30 [0.93, 1.74]	0.0%	1.35 [0.03, 1.77]	-8.9% [-10.8%, -3.0%]	-2.4% [-2.9%, -1.8%]
4	bb_pctb_20d	8.1% [4.8%, 12.0%]	1.25 [0.87, 1.72]	0.0%	1.26 [0.14, 1.81]	-7.5% [-9.8%, -3.1%]	-2.3% [-2.9%, -1.7%]
5	eps_growth_yoy	5.5% [1.8%, 9.2%]	1.07 [0.30, 2.07]	0.5%	-1.05 [-1.65, -0.06]	-11.7% [-18.6%, -3.6%]	-4.8% [-4.8%, -1.9%]
6	capex_to_revenue	4.4% [2.5%, 6.4%]	0.98 [0.55, 1.47]	0.0%	0.52 [-0.23, 1.18]	-5.6% [-9.2%, -3.9%]	-2.0% [-2.4%, -1.6%]
7	net_income_growth_yoy	4.5% [1.5%, 7.7%]	0.97 [0.28, 1.92]	0.4%	-1.44 [-2.45, 0.50]	-11.4% [-17.7%, -3.3%]	-5.0% [-5.0%, -1.5%]
8	rsi_14d	5.7% [2.5%, 9.1%]	0.92 [0.43, 1.40]	0.0%	0.77 [0.10, 1.25]	-12.7% [-15.9%, -3.2%]	-2.7% [-3.5%, -2.0%]
9	dpo	4.1% [2.2%, 5.9%]	0.92 [0.48, 1.48]	0.0%	-0.37 [-1.08, 0.80]	-8.9% [-12.7%, -3.2%]	-2.4% [-3.7%, -1.4%]
10	price_zscore_63d	5.6% [2.2%, 9.8%]	0.84 [0.36, 1.31]	0.0%	0.92 [0.32, 1.19]	-7.1% [-12.6%, -3.2%]	-2.4% [-2.7%, -2.1%]
11	debt_to_equity	4.4% [0.9%, 7.9%]	0.83 [0.17, 1.56]	0.7%	-0.12 [-0.84, 0.40]	-8.8% [-16.1%, -4.6%]	-3.0% [-3.9%, -2.0%]
12	debt_to_assets	4.1% [1.3%, 6.7%]	0.81 [0.27, 1.35]	0.2%	-0.50 [-1.61, 0.46]	-7.4% [-13.6%, -4.4%]	-2.8% [-4.1%, -1.8%]
13	gp_to_equity	4.4% [1.3%, 7.3%]	0.81 [0.26, 1.37]	0.2%	0.22 [-0.68, 0.70]	-8.7% [-15.3%, -5.0%]	-2.9% [-3.5%, -2.3%]
14	momentum_0_21	5.3% [2.1%, 8.5%]	0.81 [0.38, 1.27]	0.0%	0.99 [0.58, 1.33]	-9.7% [-15.3%, -3.9%]	-2.6% [-3.2%, -2.0%]
15	accruals_total	3.9% [1.6%, 6.3%]	0.75 [0.33, 1.31]	0.1%	1.41 [-0.79, 2.42]	-7.2% [-12.2%, -4.0%]	-2.5% [-3.3%, -1.7%]
16	momentum_21_252_vol_63d	6.5% [2.4%, 8.6%]	0.69 [0.30, 1.13]	0.0%	-0.13 [-0.46, 0.21]	-10.2% [-17.4%, -7.0%]	-4.8% [-5.5%, -3.3%]
17	accruals_noa_proxy	3.4% [0.7%, 6.2%]	0.68 [0.15, 1.24]	0.5%	0.99 [-0.33, 1.88]	-7.7% [-13.5%, -4.2%]	-2.6% [-3.1%, -1.7%]
18	momentum_21_252	5.6% [2.3%, 8.9%]	0.67 [0.26, 1.13]	0.0%	-0.15 [-0.54, 0.21]	-10.2% [-18.1%, -7.3%]	-5.1% [-5.9%, -3.6%]
19	share_turnover_21d	5.5% [-0.3%, 9.7%]	0.65 [-0.03, 1.70]	2.9%	-3.33 [-4.26, 0.68]	-31.1% [-45.8%, -3.7%]	-5.6% [-10.4%, -1.8%]
20	asset_turnover	4.1% [1.6%, 6.5%]	0.65 [0.26, 1.03]	0.1%	-0.03 [-0.31, 0.21]	-7.9% [-13.9%, -6.2%]	-3.4% [-3.9%, -2.6%]
21	ma_dist_63d	4.0% [0.6%, 8.0%]	0.53 [0.08, 0.95]	1.1%	0.92 [0.36, 1.17]	-11.1% [-20.1%, -6.4%]	-2.6% [-2.7%, -2.5%]

Risk Loadings

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ret_zscore_63d	0.03 (1.07)	0.21 (3.15)	0.07 (1.60)	-0.05 (-1.04)	0.12 (1.06)	0.07 (0.81)	6.91% (4.89)	0.188
resid_ret_252v_1260c_SPY	0.04 (1.03)	0.19 (2.26)	0.07 (1.39)	-0.07 (-1.44)	0.14 (1.08)	-0.05 (-0.52)	7.70% (4.21)	0.213
stoch_k_14d	0.12 (2.44)	0.25 (3.97)	0.04 (0.64)	-0.11 (-1.53)	0.03 (0.33)	-0.09 (-1.30)	8.49% (4.89)	0.335
bb_pctb_20d	0.13 (2.67)	0.15 (2.65)	0.03 (0.37)	-0.11 (-1.56)	0.01 (0.09)	-0.02 (-0.36)	7.23% (4.11)	0.250
eps_growth_yoy	0.04 (1.27)	-0.05 (-0.89)	0.11 (2.36)	0.23 (4.68)	0.08 (1.57)	0.07 (0.87)	2.88% (1.71)	0.243
capex_to_revenue	0.01 (0.50)	0.27 (4.01)	-0.03 (-0.90)	0.00 (0.04)	0.06 (1.33)	0.25 (4.95)	2.47% (1.96)	0.264
net_income_growth_yoy	0.03 (0.85)	-0.06 (-1.55)	0.10 (3.01)	0.19 (4.82)	0.07 (1.51)	0.15 (2.05)	1.96% (1.36)	0.313
rsi_14d	0.11 (2.64)	0.13 (2.41)	0.01 (0.20)	-0.11 (-1.48)	-0.10 (-1.42)	-0.05 (-1.05)	6.15% (3.38)	0.264
dpo	0.03 (0.87)	0.10 (1.54)	-0.07 (-2.40)	0.03 (1.09)	0.04 (0.62)	0.18 (2.27)	2.08% (1.38)	0.158
price_zscore_63d	0.09 (2.27)	0.02 (0.27)	-0.03 (-0.71)	-0.26 (-5.60)	-0.10 (-1.54)	-0.11 (-1.87)	7.15% (4.23)	0.368
debt_to_equity	0.09 (2.86)	-0.33 (-6.93)	0.14 (3.49)	-0.01 (-0.33)	0.14 (2.33)	-0.06 (-1.11)	2.47% (1.95)	0.450
debt_to_assets	0.06 (1.76)	-0.15 (-2.62)	0.16 (3.62)	0.02 (0.43)	0.15 (2.20)	-0.10 (-1.54)	2.73% (2.18)	0.261
gp_to_equity	0.05 (1.82)	-0.18 (-4.54)	-0.08 (-2.75)	-0.04 (-1.52)	0.14 (3.48)	0.32 (6.17)	0.51% (0.37)	0.498
momentum_0_21	0.15 (2.74)	0.03 (0.53)	-0.00 (-0.05)	-0.19 (-3.52)	-0.05 (-0.90)	-0.09 (-1.47)	5.38% (3.61)	0.365
accruals_total	-0.10 (-3.26)	0.19 (3.70)	0.02 (0.40)	0.01 (0.25)	0.12 (1.72)	-0.27 (-4.25)	5.81% (4.03)	0.395
momentum_21_252_vol_63d	0.03 (0.98)	-0.12 (-2.33)	-0.05 (-1.20)	0.49 (9.65)	0.17 (3.37)	-0.05 (-0.60)	1.12% (0.84)	0.680
accruals_noa_proxy	-0.11 (-3.11)	0.07 (1.16)	0.20 (3.56)	0.06 (1.39)	0.05 (0.41)	-0.06 (-0.80)	4.94% (3.57)	0.265
momentum_21_252	0.02 (0.77)	-0.06 (-1.11)	-0.04 (-0.70)	0.53 (9.02)	0.19 (3.21)	-0.09 (-1.65)	1.31% (1.03)	0.679
share_turnover_21d	-0.01 (-0.23)	-0.31 (-2.63)	-0.00 (-0.05)	-0.03 (-0.54)	-0.27 (-1.48)	0.44 (3.61)	5.47% (2.97)	0.469
asset_turnover	0.08 (2.45)	0.17 (3.03)	0.03 (0.75)	-0.01 (-0.20)	0.08 (1.35)	0.59 (10.05)	-0.77% (-0.56)	0.487
ma_dist_63d	0.13 (2.92)	-0.05 (-0.80)	-0.03 (-0.77)	-0.31 (-6.49)	-0.17 (-3.22)	-0.14 (-2.15)	6.12% (3.74)	0.462

Return Contributions

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ret_zscore_63d	0.44%	-0.18%	-0.17%	-0.24%	1.10%	0.39%	6.91%	8.24%
resid_ret_252v_1260c_SPY	0.47%	-0.17%	-0.17%	-0.37%	1.24%	-0.30%	7.70%	8.40%
stoch_k_14d	1.58%	-0.22%	-0.10%	-0.53%	0.26%	-0.53%	8.49%	8.96%
bb_pctb_20d	1.71%	-0.13%	-0.06%	-0.57%	0.07%	-0.11%	7.23%	8.13%
eps_growth_yoy	0.55%	0.04%	-0.26%	1.15%	0.75%	0.38%	2.88%	5.49%
capex_to_revenue	0.16%	-0.23%	0.06%	0.01%	0.51%	1.45%	2.47%	4.42%
net_income_growth_yoy	0.33%	0.05%	-0.24%	0.95%	0.66%	0.84%	1.96%	4.55%
rsi_14d	1.42%	-0.11%	-0.03%	-0.56%	-0.85%	-0.30%	6.15%	5.72%
dpo	0.34%	-0.09%	0.15%	0.17%	0.35%	1.04%	2.08%	4.05%
price_zscore_63d	1.18%	-0.02%	0.08%	-1.31%	-0.90%	-0.60%	7.15%	5.57%
debt_to_equity	1.10%	0.29%	-0.33%	-0.07%	1.22%	-0.31%	2.47%	4.36%
debt_to_assets	0.70%	0.13%	-0.36%	0.08%	1.32%	-0.55%	2.73%	4.05%
gp_to_equity	0.69%	0.16%	0.18%	-0.23%	1.28%	1.81%	0.51%	4.40%
momentum_0_21	1.87%	-0.03%	0.01%	-0.98%	-0.48%	-0.51%	5.38%	5.26%
accruals_total	-1.25%	-0.17%	-0.04%	0.05%	1.07%	-1.55%	5.81%	3.92%
momentum_21_252_vol_63d	0.40%	0.11%	0.12%	2.49%	1.55%	-0.26%	1.12%	5.52%
accruals_noa_proxy	-1.44%	-0.06%	-0.45%	0.32%	0.40%	-0.33%	4.94%	3.38%
momentum_21_252	0.30%	0.05%	0.09%	2.68%	1.69%	-0.53%	1.31%	5.60%
share_turnover_21d	-0.16%	0.27%	0.01%	-0.17%	-2.39%	2.52%	5.47%	5.54%
asset_turnover	1.03%	-0.15%	-0.07%	-0.04%	0.68%	3.37%	-0.77%	4.05%
ma_dist_63d	1.66%	0.05%	0.07%	-1.57%	-1.53%	-0.80%	6.12%	4.00%

1. ret_zscore_63d

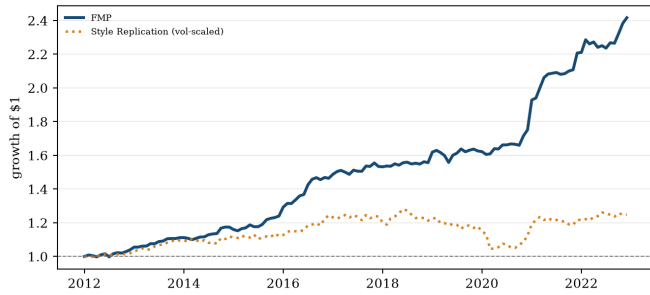


Figure: Common-window cumulative growth of \$1.

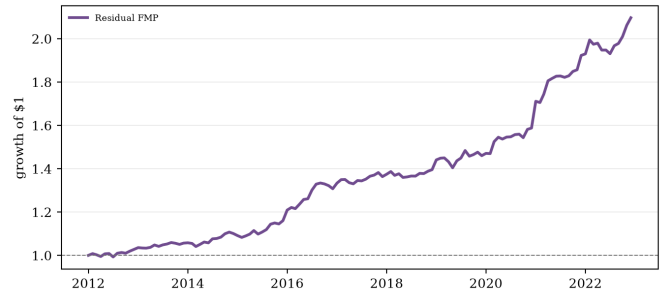


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ret_zscore_63d	8.24%	5.21%	1.58	4.51	-4.40%	0.179	7.32%	5.21%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ret_zscore_63d	0.03 (1.07)	0.21 (3.15)	0.07 (1.60)	-0.05 (-1.04)	0.12 (1.06)	0.07 (0.81)	6.91% (4.89)	0.188

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ret_zscore_63d	0.44%	-0.18%	-0.17%	-0.24%	1.10%	0.39%	6.91%	8.24%

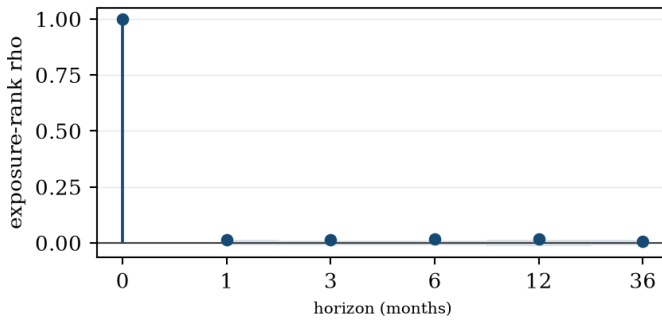


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

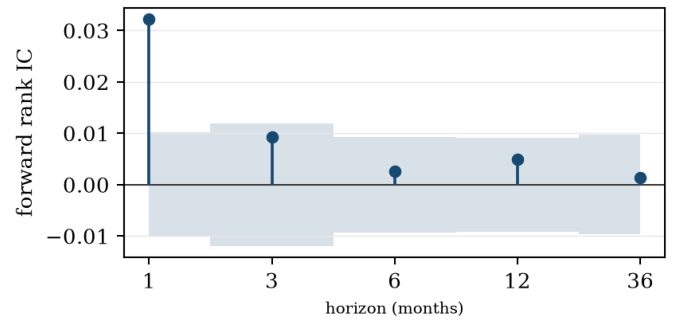


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	ret_zscore_63d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1267 / 1582 / 2103
Available-history Sharpe	1.58
Bootstrap Sharpe	1.58 [1.14, 2.06]; Pr(S<=0)=0.0%
Bootstrap tails	skew=2.29 [0.50, 3.05]; max DD=-4.4% [-5.5%, -1.9%]

2. resid_ret_252v_1260c_SPY

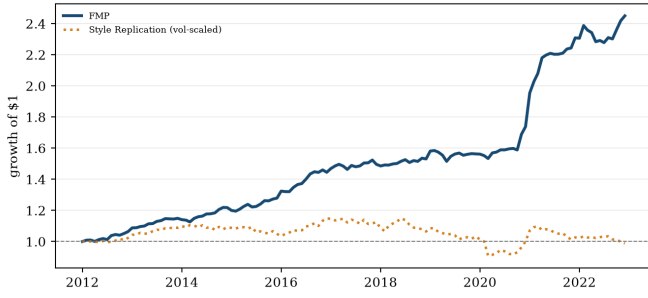


Figure: Common-window cumulative growth of \$1.

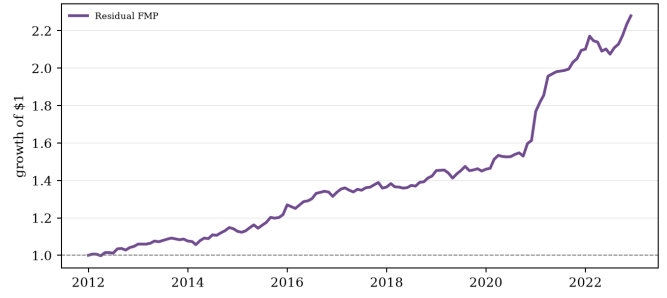


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
resid_ret_252v_1260c_SPY	8.40%	5.88%	1.43	3.53	-4.59%	0.197	8.37%	5.88%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
resid_ret_252v_1260c_SPY	0.04 (1.03)	0.19 (2.26)	0.07 (1.39)	-0.07 (-1.44)	0.14 (1.08)	-0.05 (-0.52)	7.70% (4.21)	0.213

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
resid_ret_252v_1260c_SPY	0.47%	-0.17%	-0.17%	-0.37%	1.24%	-0.30%	7.70%	8.40%

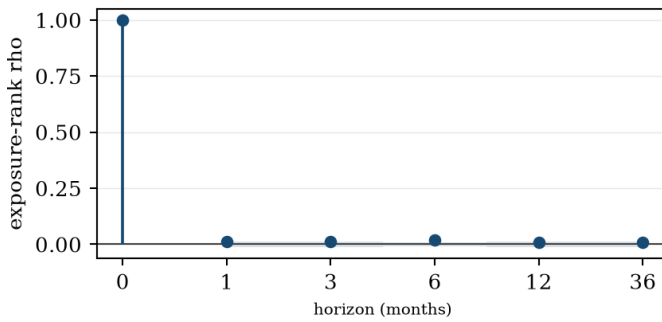


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

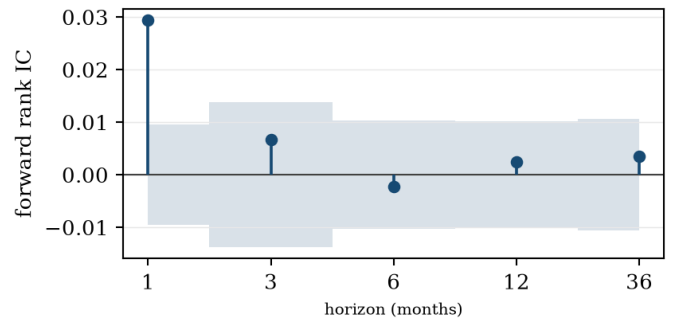


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	resid_ret_252v_1260c_SPY(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	1.43
Bootstrap Sharpe	1.43 [1.04, 1.97]; Pr(S<=0)=0.0%
Bootstrap tails	skew=2.96 [-0.03, 3.51]; max DD=-4.6% [-7.1%, -2.3%]

3. stoch_k_14d

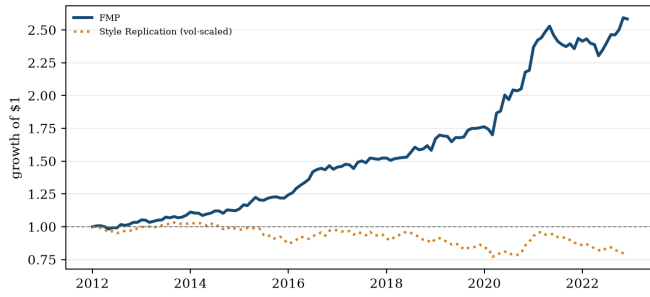


Figure: Common-window cumulative growth of \$1.

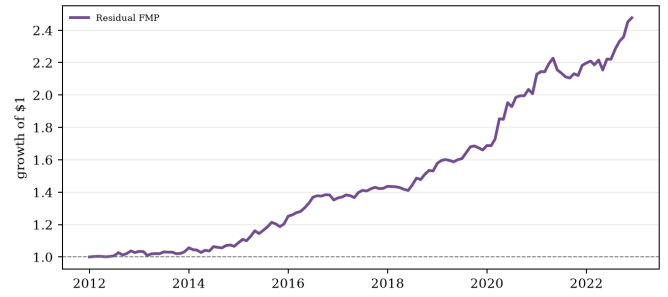


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
stoch_k_14d	8.96%	6.90%	1.30	4.22	-8.91%	0.294	10.04%	6.90%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
stoch_k_14d	0.12 (2.44)	0.25 (3.97)	0.04 (0.64)	-0.11 (-1.53)	0.03 (0.33)	-0.09 (-1.30)	8.49% (4.89)	0.335

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
stoch_k_14d	1.58%	-0.22%	-0.10%	-0.53%	0.26%	-0.53%	8.49%	8.96%

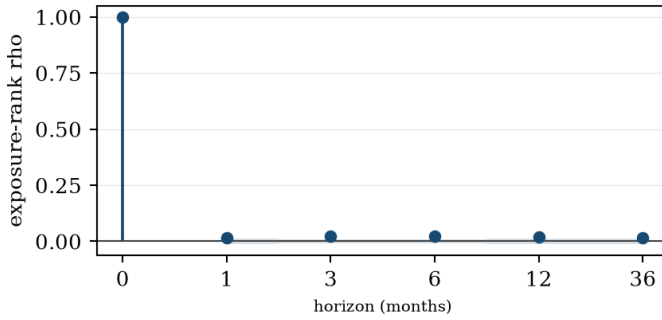


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

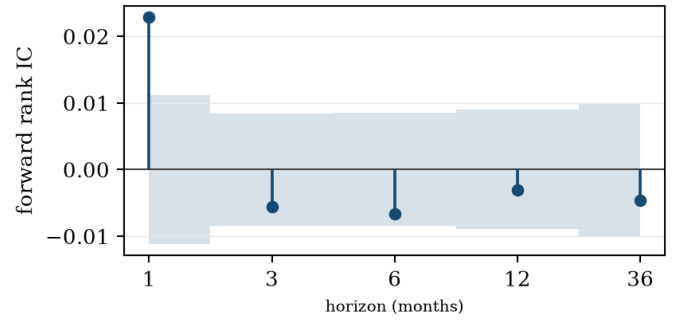


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	stoch_k_14d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1266 / 1578 / 2103
Available-history Sharpe	1.30
Bootstrap Sharpe	1.30 [0.93, 1.74]; Pr(S<=0)=0.0%
Bootstrap tails	skew=1.35 [0.03, 1.77]; max DD=-8.9% [-10.8%, -3.0%]

4. bb_pctb_20d

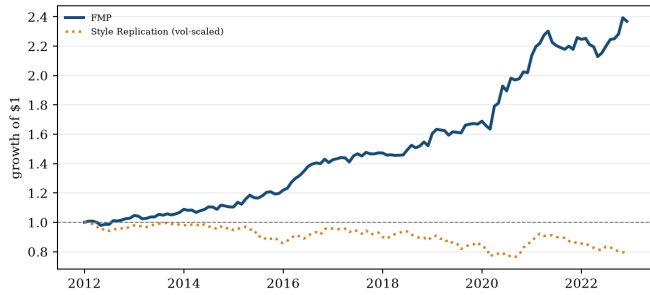


Figure: Common-window cumulative growth of \$1.

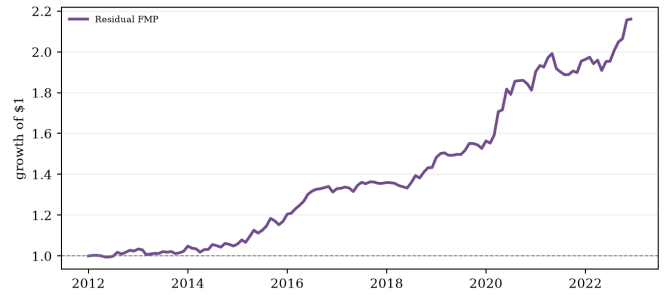


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
bb_pctb_20d	8.13%	6.49%	1.25	4.20	-7.53%	0.189	9.00%	6.49%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
bb_pctb_20d	0.13 (2.67)	0.15 (2.65)	0.03 (0.37)	-0.11 (-1.56)	0.01 (0.09)	-0.02 (-0.36)	7.23% (4.11)	0.250

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
bb_pctb_20d	1.71%	-0.13%	-0.06%	-0.57%	0.07%	-0.11%	7.23%	8.13%

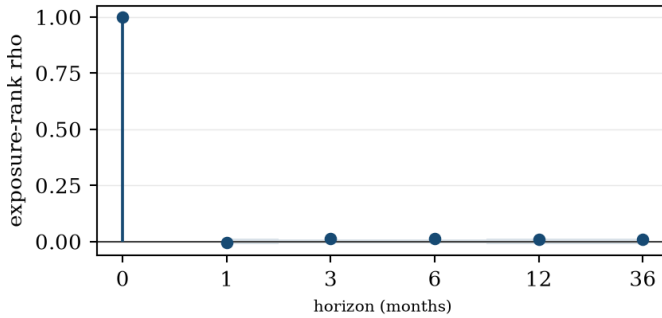


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

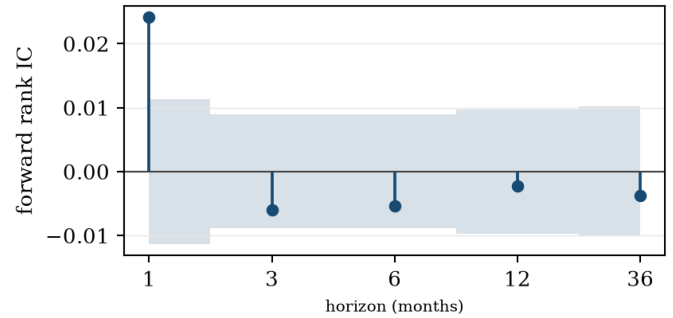


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	bb_pctb_20d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1266 / 1579 / 2103
Available-history Sharpe	1.25
Bootstrap Sharpe	1.25 [0.87, 1.72]; Pr(S<=0)=0.0%
Bootstrap tails	skew=1.26 [0.14, 1.81]; max DD=-7.5% [-9.8%, -3.1%]

5. eps_growth_yoy

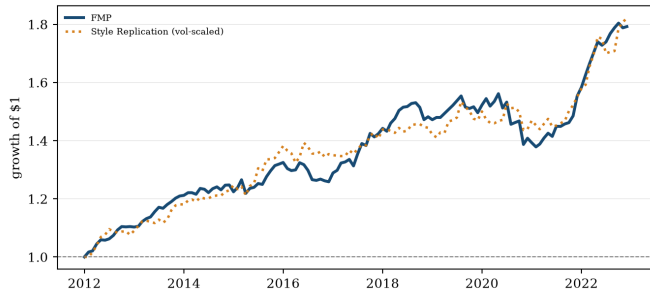


Figure: Common-window cumulative growth of \$1.

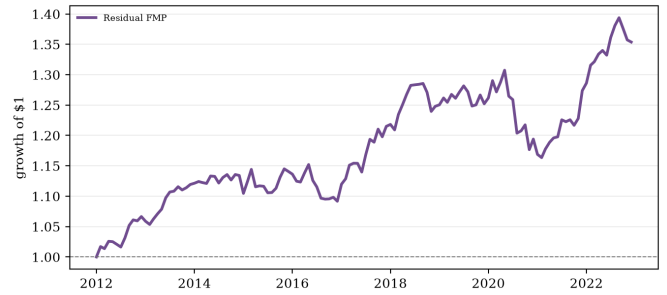


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
eps_growth_yoy	5.49%	5.14%	1.07	3.04	-11.71%	0.241	2.72%	5.14%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
eps_growth_yoy	0.04 (1.27)	-0.05 (-0.89)	0.11 (2.36)	0.23 (4.68)	0.08 (1.57)	0.07 (0.87)	2.88% (1.71)	0.243

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
eps_growth_yoy	0.55%	0.04%	-0.26%	1.15%	0.75%	0.38%	2.88%	5.49%

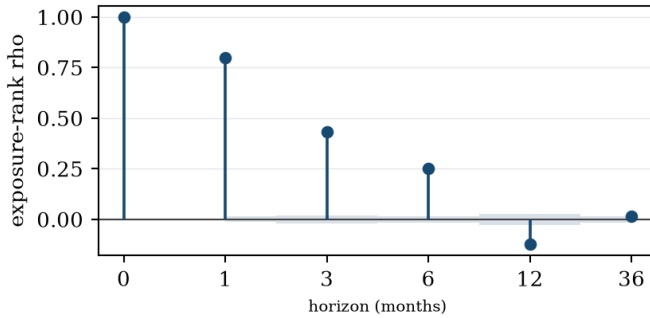


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

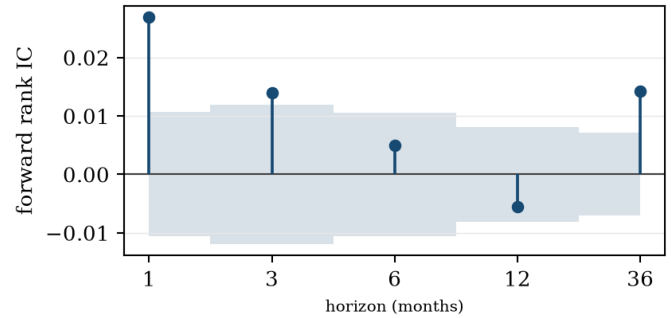


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	eps_growth_yoy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1235 / 1540 / 2060
Available-history Sharpe	1.07
Bootstrap Sharpe	1.07 [0.30, 2.07]; Pr(S<=0)=0.5%
Bootstrap tails	skew=-1.05 [-1.65, -0.06]; max DD=-11.7% [-18.6%, -3.8%]

6. capex_to_revenue

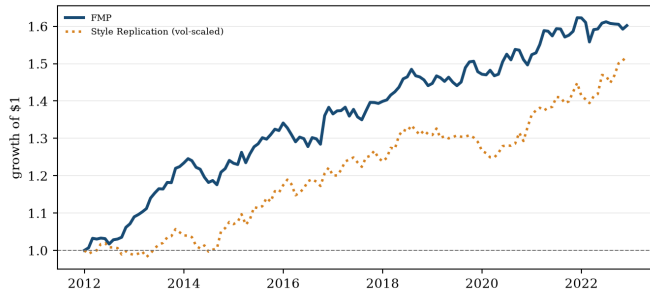


Figure: Common-window cumulative growth of \$1.

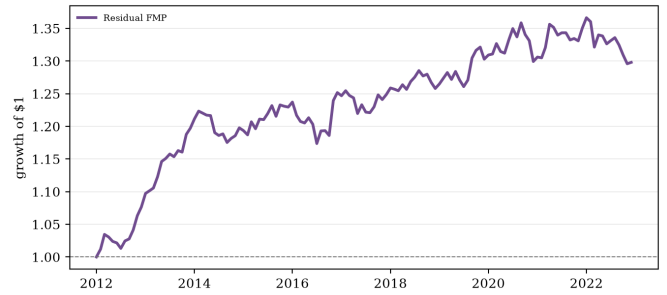


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
capex_to_revenue	4.42%	4.53%	0.98	3.38	-5.61%	0.263	2.40%	4.53%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
capex_to_revenue	0.01 (0.50)	0.27 (4.01)	-0.03 (-0.90)	0.00 (0.04)	0.06 (1.33)	0.25 (4.95)	2.47% (1.96)	0.264

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
capex_to_revenue	0.16%	-0.23%	0.06%	0.01%	0.51%	1.45%	2.47%	4.42%

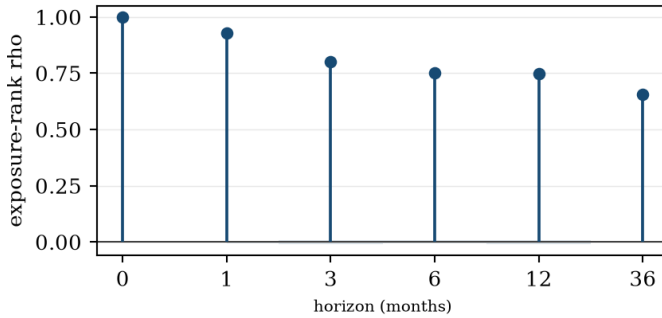


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

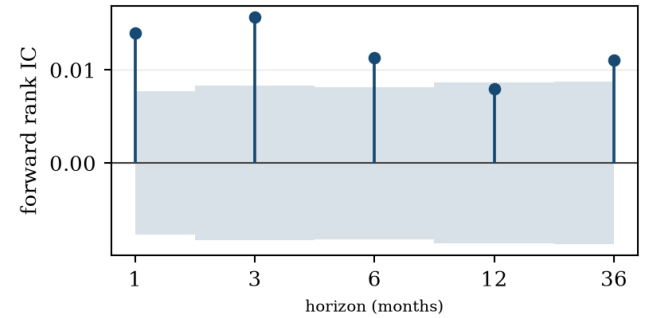


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	capex_to_revenue(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1258 / 1556 / 2035
Available-history Sharpe	0.98
Bootstrap Sharpe	0.98 [0.55, 1.47]; Pr(S<=0)=0.0%
Bootstrap tails	skew=0.52 [-0.23, 1.18]; max DD=-5.6% [-9.2%, -3.9%]

7. net_income_growth_yoy

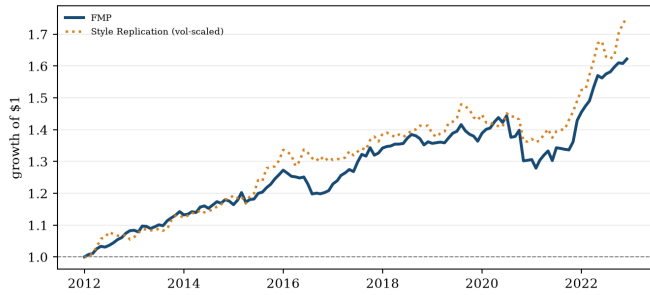


Figure: Common-window cumulative growth of \$1.

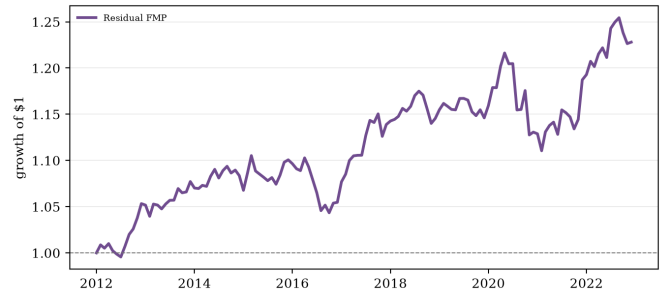


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
net_income_growth_yoy	4.55%	4.67%	0.97	2.88	-11.44%	0.309	1.63%	4.67%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
net_income_growth_yoy	0.03	-0.06	0.10	0.19	0.07	0.15	1.96%	0.313
	(0.85)	(-1.55)	(3.01)	(4.82)	(1.51)	(2.05)	(1.36)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
net_income_growth_yoy	0.33%	0.05%	-0.24%	0.95%	0.66%	0.84%	1.96%	4.55%

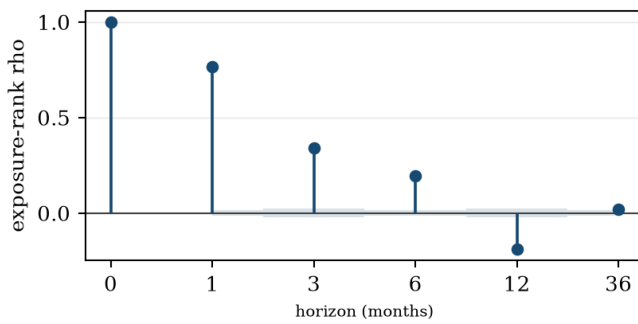


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

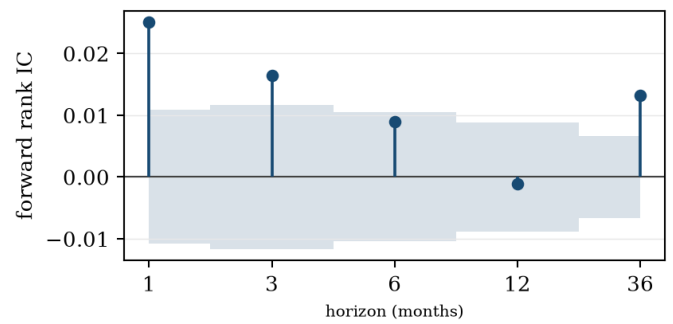


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	net_income_growth_yoy(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1259 / 1565 / 2085
Available-history Sharpe	0.97
Bootstrap Sharpe	0.97 [0.28, 1.92]; Pr(S<=0)=0.4%
Bootstrap tails	skew=-1.44 [-2.45, 0.50]; max DD=-11.4% [-17.7%, -3.5%]

8. rsi_14d

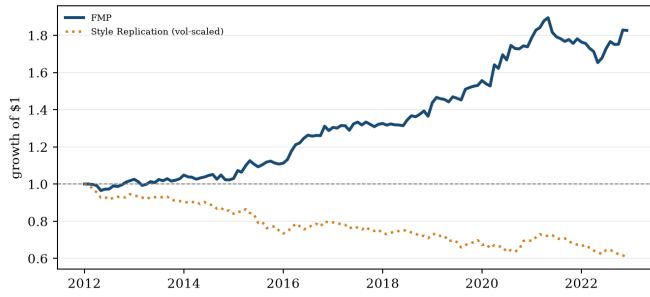


Figure: Common-window cumulative growth of \$1.

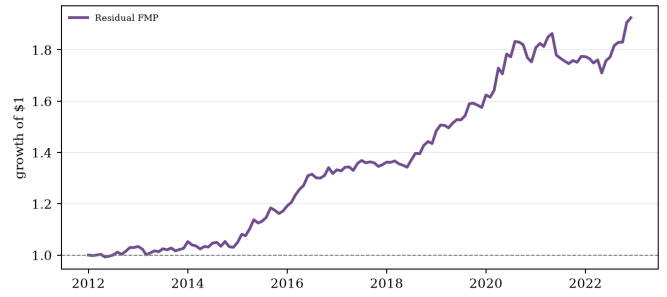


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
rsi_14d	5.72%	6.20%	0.92	3.26	-12.74%	0.220	7.76%	6.20%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
rsi_14d	0.11 (2.64)	0.13 (2.41)	0.01 (0.20)	-0.11 (-1.48)	-0.10 (-1.42)	-0.05 (-1.05)	6.15% (3.38)	0.264

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
rsi_14d	1.42%	-0.11%	-0.03%	-0.56%	-0.85%	-0.30%	6.15%	5.72%

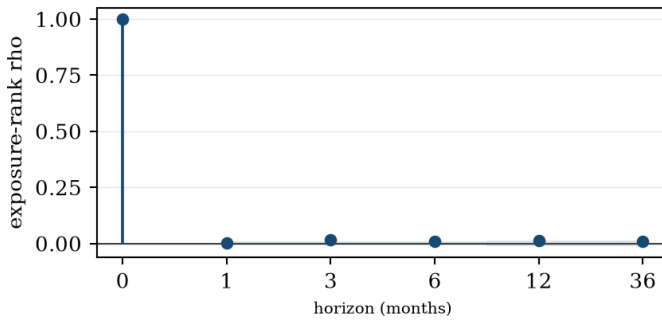


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

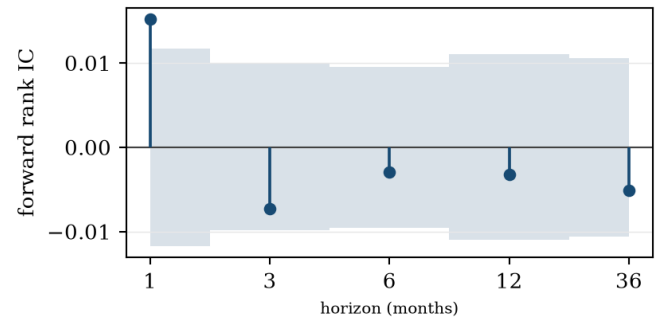


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	rsi_14d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1266 / 1578 / 2103
Available-history Sharpe	0.92
Bootstrap Sharpe	0.92 [0.43, 1.40]; Pr(S<=0)=0.0%
Bootstrap tails	skew=0.77 [0.10, 1.25]; max DD=-12.7% [-15.9%, -3.2%]

9. dpo

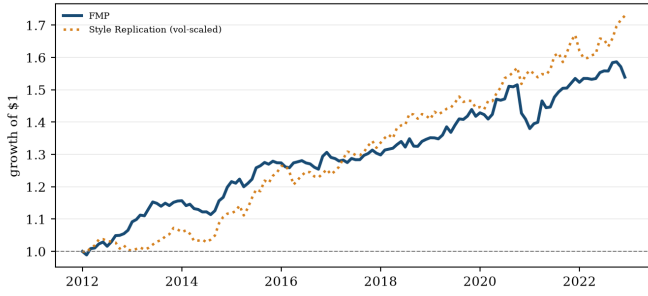


Figure: Common-window cumulative growth of \$1.

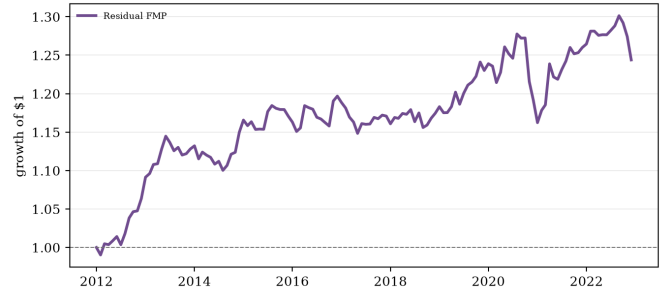


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
dpo	4.05%	4.43%	0.92	3.03	-8.94%	0.158	2.02%	4.43%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
dpo	0.03 (0.87)	0.10 (1.54)	-0.07 (-2.40)	0.03 (1.09)	0.04 (0.62)	0.18 (2.27)	2.08% (1.38)	0.158

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
dpo	0.34%	-0.09%	0.15%	0.17%	0.35%	1.04%	2.08%	4.05%

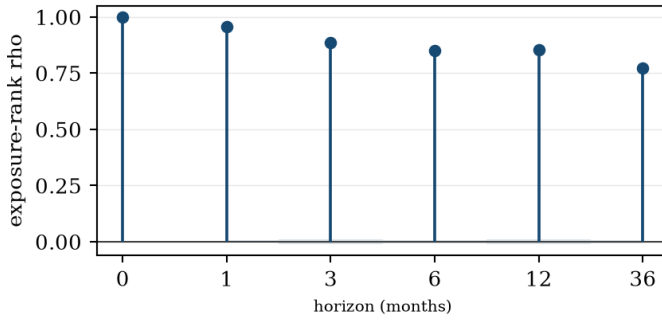


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

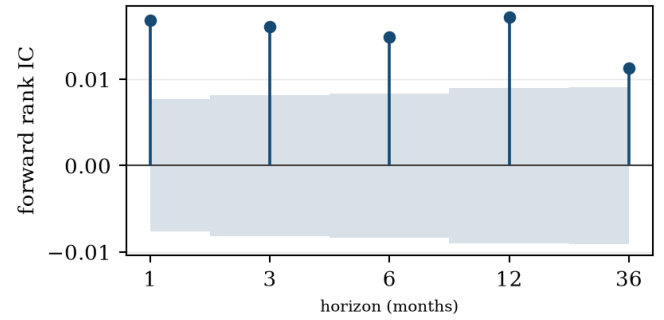


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	dpo(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1208 / 1548 / 2016
Available-history Sharpe	0.92
Bootstrap Sharpe	0.92 [0.48, 1.48]; Pr(S<=0)=0.0%
Bootstrap tails	skew=-0.37 [-1.08, 0.80]; max DD=-8.9% [-12.7%, -3.2%]

10. price_zscore_63d

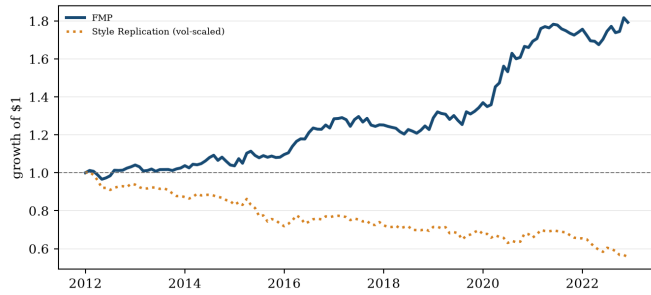


Figure: Common-window cumulative growth of \$1.

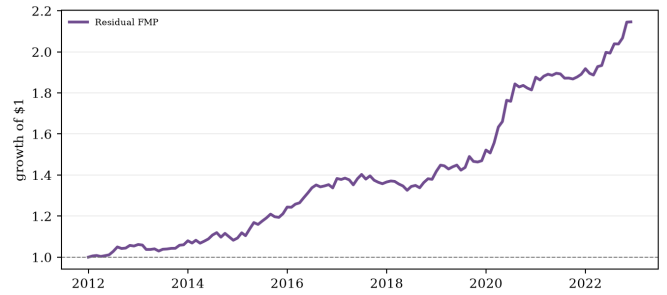


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
price_zscore_63d	5.57%	6.65%	0.84	2.83	-7.14%	0.340	8.57%	6.65%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
price_zscore_63d	0.09 (2.27)	0.02 (0.27)	-0.03 (-0.71)	-0.26 (-5.60)	-0.10 (-1.54)	-0.11 (-1.87)	7.15% (4.23)	0.368

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
price_zscore_63d	1.18%	-0.02%	0.08%	-1.31%	-0.90%	-0.60%	7.15%	5.57%

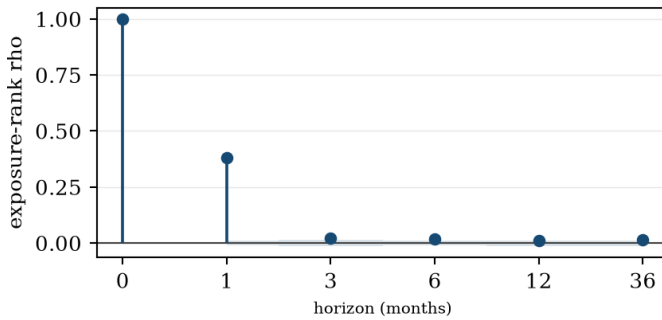


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

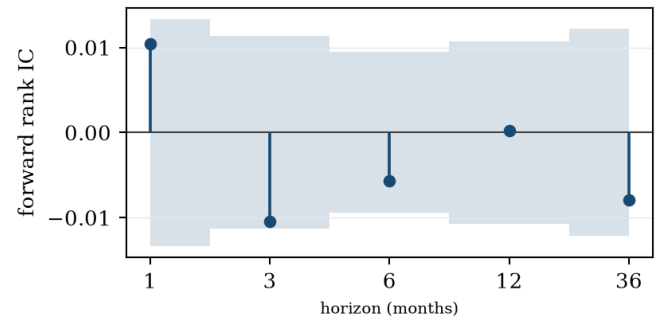


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	price_zscore_63d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1267 / 1582 / 2103
Available-history Sharpe	0.84
Bootstrap Sharpe	0.84 [0.36, 1.31]; Pr(S<=0)=0.0%
Bootstrap tails	skew=0.92 [0.32, 1.19]; max DD=-7.1% [-12.6%, -5.1%]

11. debt_to_equity

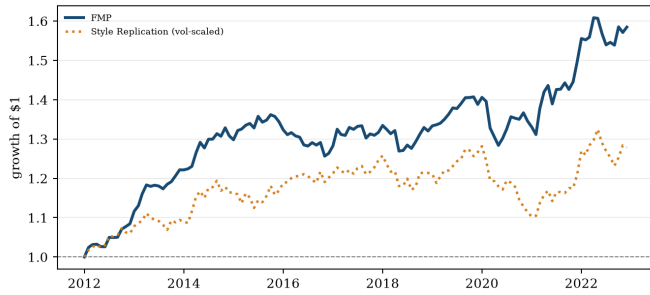


Figure: Common-window cumulative growth of \$1.

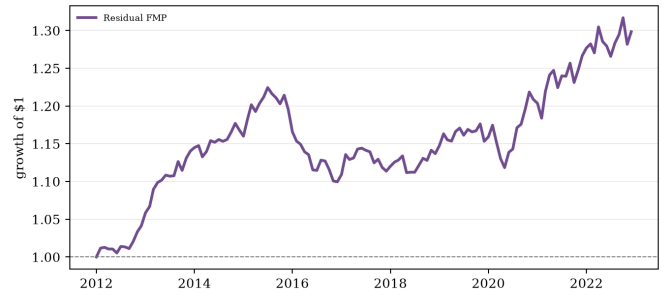


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
debt_to_equity	4.36%	5.28%	0.83	2.62	-8.76%	0.445	2.77%	5.28%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
debt_to_equity	0.09 (2.86)	-0.33 (-6.93)	0.14 (3.49)	-0.01 (-0.33)	0.14 (2.33)	-0.06 (-1.11)	2.47% (1.95)	0.450

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
debt_to_equity	1.10%	0.29%	-0.33%	-0.07%	1.22%	-0.31%	2.47%	4.36%

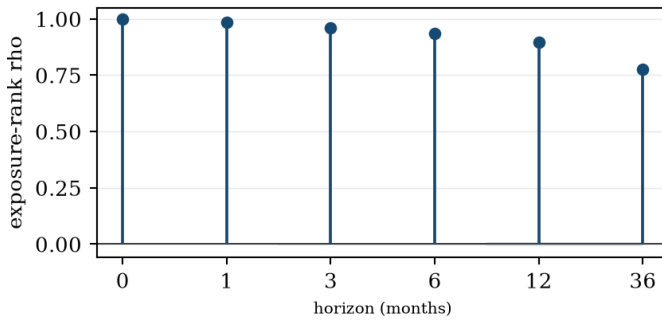


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

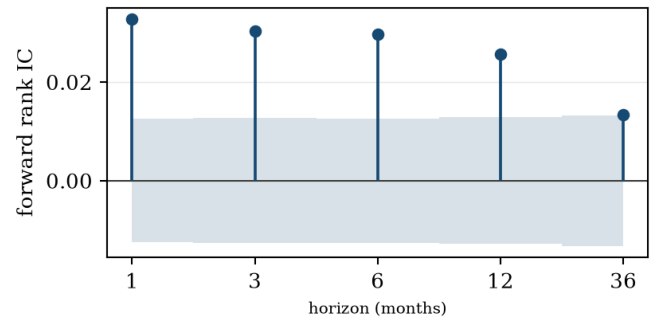


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	debt_to_equity(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1229 / 1513 / 1982
Available-history Sharpe	0.83
Bootstrap Sharpe	0.83 [0.17, 1.56]; Pr(S<=0)=0.7%
Bootstrap tails	skew=-0.12 [-0.84, 0.40]; max DD=-8.8% [-16.1%, -4.6%]

12. debt_to_assets

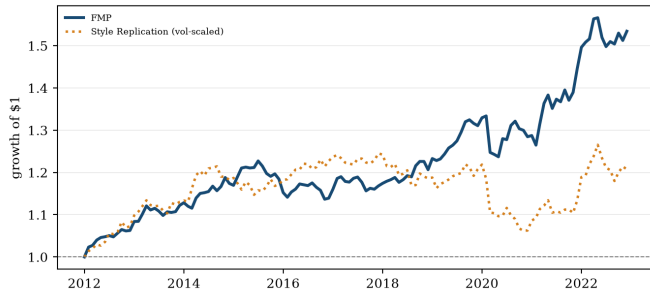


Figure: Common-window cumulative growth of \$1.

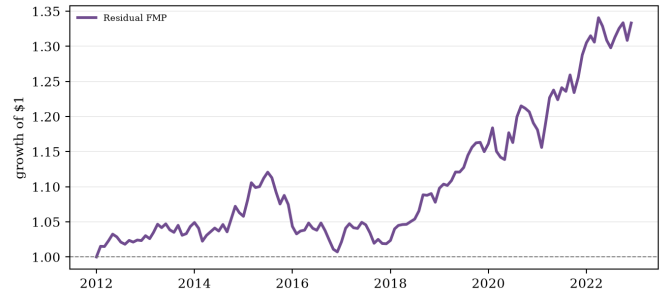


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
debt_to_assets	4.05%	4.97%	0.81	2.68	-7.38%	0.252	3.12%	4.97%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
debt_to_assets	0.06 (1.76)	-0.15 (-2.62)	0.16 (3.62)	0.02 (0.43)	0.15 (2.20)	-0.10 (-1.54)	2.73% (2.18)	0.261

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
debt_to_assets	0.70%	0.13%	-0.36%	0.08%	1.32%	-0.55%	2.73%	4.05%

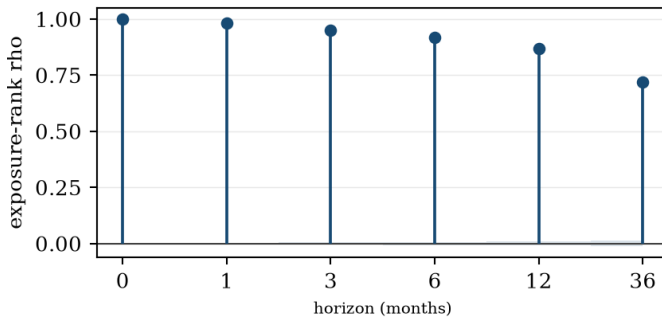


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

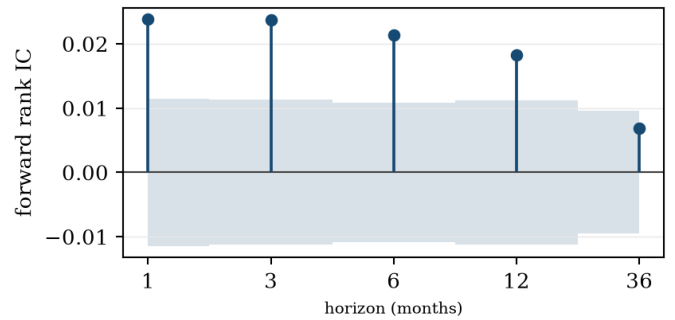


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	debt_to_assets(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1071 / 1381 / 2061
Available-history Sharpe	0.81
Bootstrap Sharpe	0.81 [0.27, 1.35]; Pr(S<=0)=0.2%
Bootstrap tails	skew=-0.50 [-1.61, 0.46]; max DD=-7.4% [-13.6%, -4.4%]

13. gp_to_equity

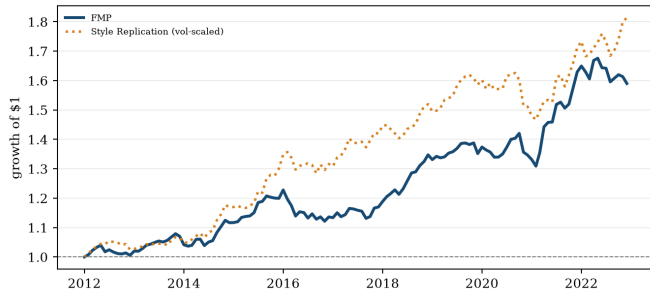


Figure: Common-window cumulative growth of \$1.



Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
gp_to_equity	4.40%	5.41%	0.81	2.51	-8.67%	0.497	0.44%	5.41%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
gp_to_equity	0.05 (1.82)	-0.18 (-4.54)	-0.08 (-2.75)	-0.04 (-1.52)	0.14 (3.48)	0.32 (6.17)	0.51% (0.37)	0.498

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
gp_to_equity	0.69%	0.16%	0.18%	-0.23%	1.28%	1.81%	0.51%	4.40%

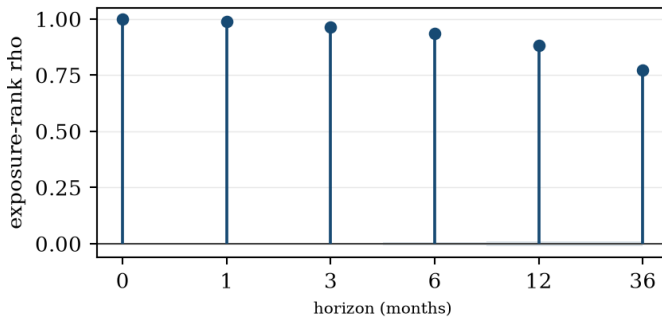


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

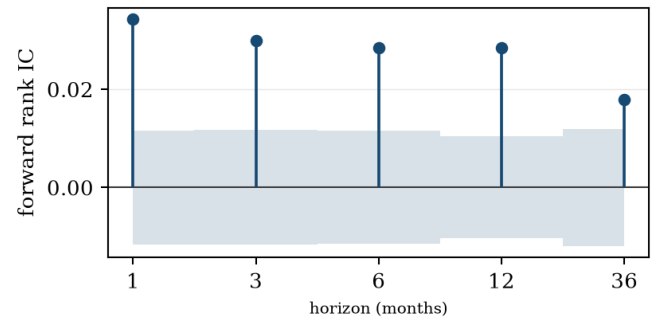


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	gp_to_equity(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1210 / 1505 / 1965
Available-history Sharpe	0.81
Bootstrap Sharpe	0.81 [0.26, 1.37]; Pr(S<=0)=0.2%
Bootstrap tails	skew=0.22 [-0.68, 0.70]; max DD=-8.7% [-15.3%, -5.0%]

14. momentum_0_21

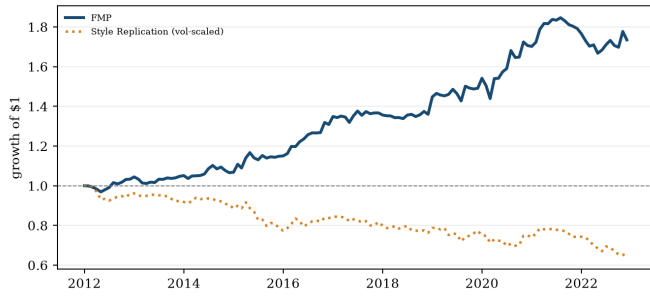


Figure: Common-window cumulative growth of \$1.

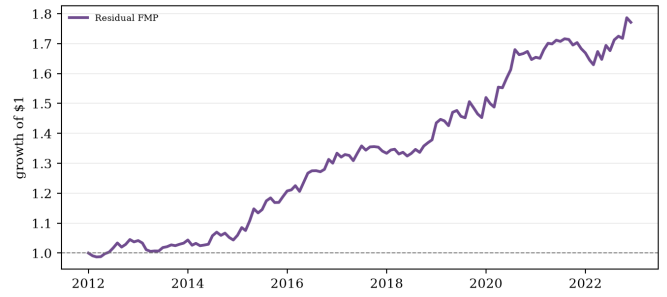


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_0_21	5.26%	6.47%	0.81	3.05	-9.68%	0.309	7.40%	6.47%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_0_21	0.15 (2.74)	0.03 (0.53)	-0.00 (-0.05)	-0.19 (-3.52)	-0.05 (-0.90)	-0.09 (-1.47)	5.38% (3.61)	0.365

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_0_21	1.87%	-0.03%	0.01%	-0.98%	-0.48%	-0.51%	5.38%	5.26%

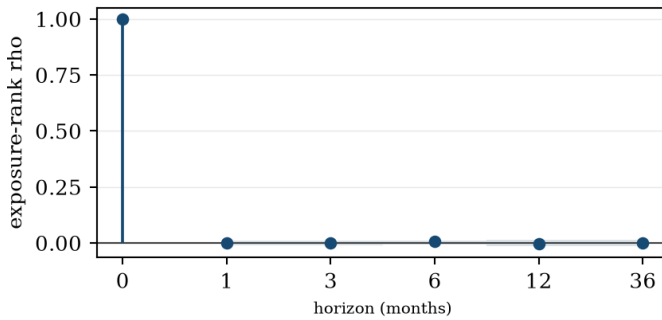


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

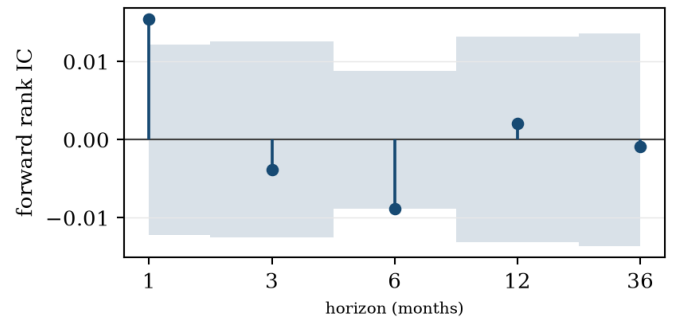


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	momentum_0_21(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.81
Bootstrap Sharpe	0.81 [0.38, 1.27]; Pr(S<=0)=0.0%
Bootstrap tails	skew=0.99 [0.58, 1.33]; max DD=-9.7% [-15.3%, -3.9%]

15. accruals_total

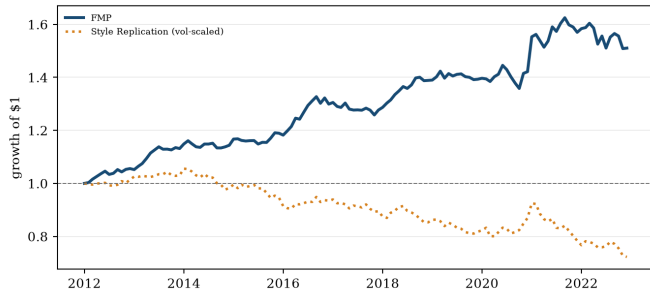


Figure: Common-window cumulative growth of \$1.

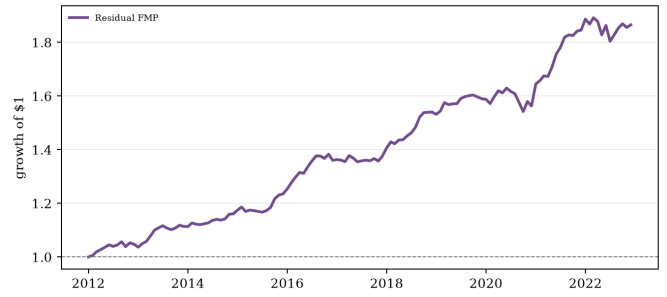


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals_total	3.92%	5.23%	0.75	2.62	-7.17%	0.394	5.70%	5.23%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals_total	-0.10 (-3.26)	0.19 (3.70)	0.02 (0.40)	0.01 (0.25)	0.12 (1.72)	-0.27 (-4.25)	5.81% (4.03)	0.395

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_total	-1.25%	-0.17%	-0.04%	0.05%	1.07%	-1.55%	5.81%	3.92%

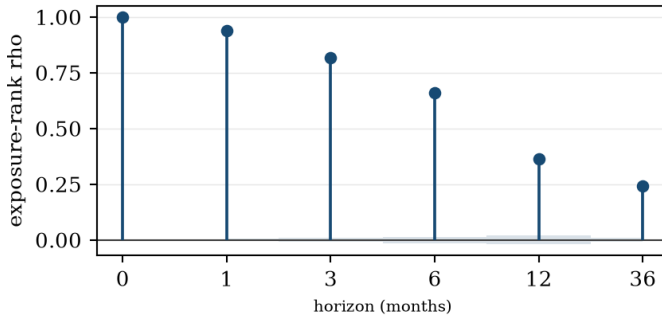


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

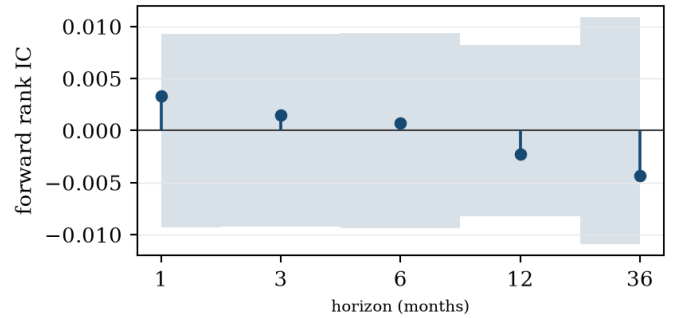


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	accruals_total(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1231 / 1556 / 2060
Available-history Sharpe	0.75
Bootstrap Sharpe	0.75 [0.33, 1.31]; Pr(S<=0)=0.1%
Bootstrap tails	skew=1.41 [-0.79, 2.42]; max DD=-7.2% [-12.2%, -4.0%]

16. momentum_21_252_vol_63d

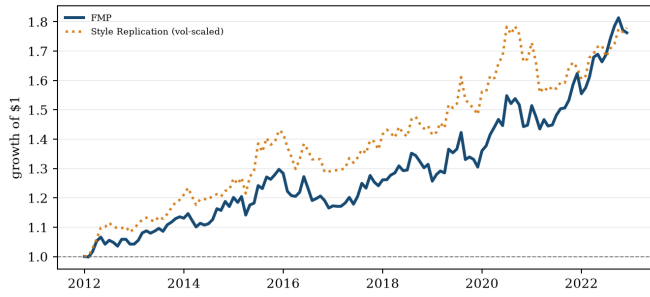


Figure: Common-window cumulative growth of \$1.

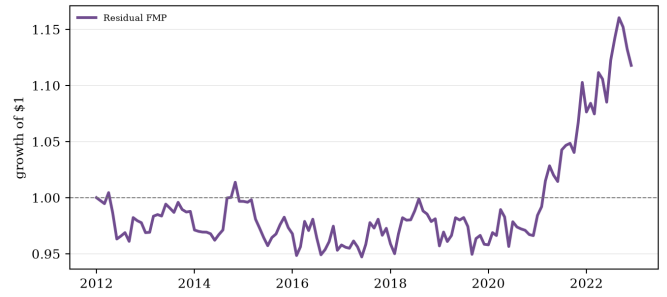


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_21_252_vol_63d	5.52%	7.95%	0.69	2.81	-10.16%	0.679	0.90%	7.95%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_21_252_vol_63d	0.03	-0.12	-0.05	0.49	0.17	-0.05	1.12%	0.680
	(0.98)	(-2.33)	(-1.20)	(9.65)	(3.37)	(-0.60)	(0.84)	

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_21_252_vol_63d	0.40%	0.11%	0.12%	2.49%	1.55%	-0.26%	1.12%	5.52%

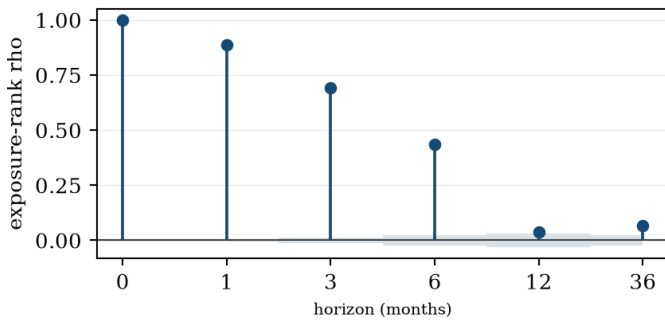


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

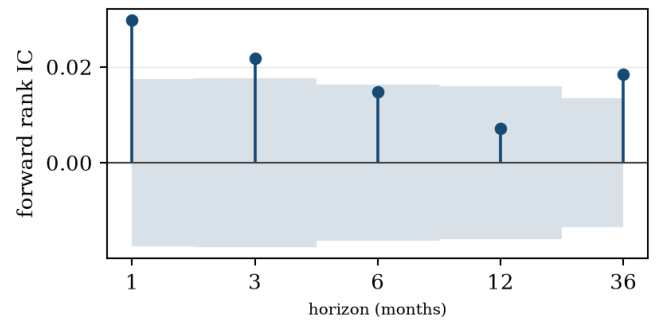


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	momentum_21_252_vol_63d(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1267 / 1582 / 2103
Available-history Sharpe	0.69
Bootstrap Sharpe	0.69 [0.30, 1.13]; Pr(S<=0)=0.0%
Bootstrap tails	skew=-0.13 [-0.46, 0.21]; max DD=-10.2% [-17.4%, -7.0%]

17. accruals_noa_proxy

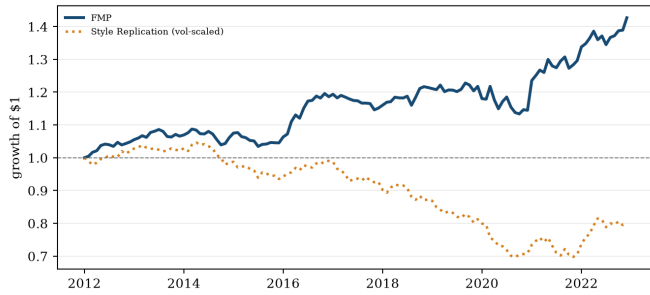


Figure: Common-window cumulative growth of \$1.

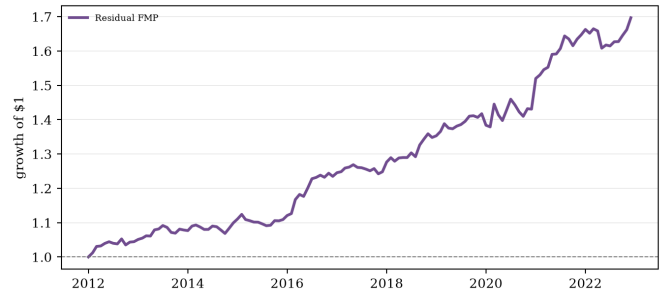


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
accruals_noa_proxy	3.38%	4.99%	0.68	2.19	-7.68%	0.241	4.32%	4.99%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
accruals_noa_proxy	-0.11 (-3.11)	0.07 (1.16)	0.20 (3.56)	0.06 (1.39)	0.05 (0.41)	-0.06 (-0.80)	4.94% (3.57)	0.265

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
accruals_noa_proxy	-1.44%	-0.06%	-0.45%	0.32%	0.40%	-0.33%	4.94%	3.38%

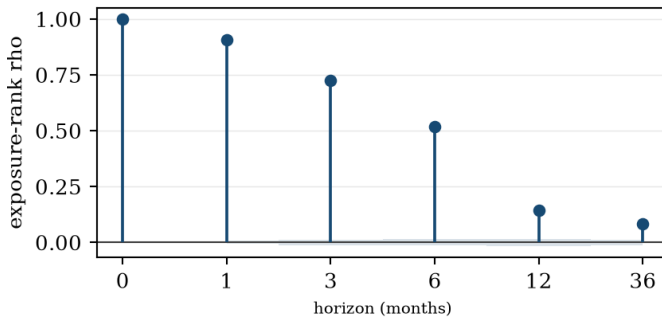


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

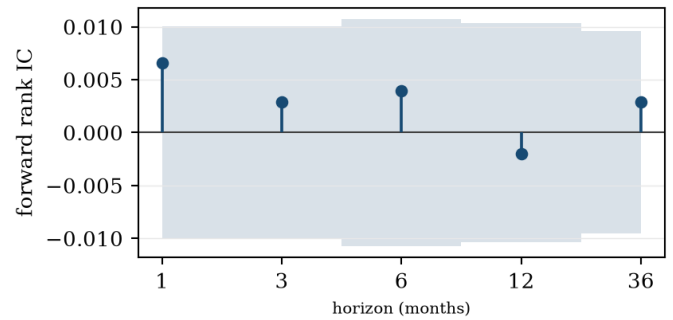


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	accruals_noa_proxy(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1033 / 1338 / 2031
Available-history Sharpe	0.68
Bootstrap Sharpe	0.68 [0.15, 1.24]; Pr(S<=0)=0.5%
Bootstrap tails	skew=0.99 [-0.33, 1.88]; max DD=-7.7% [-13.5%, -4.2%]

18. momentum_21_252

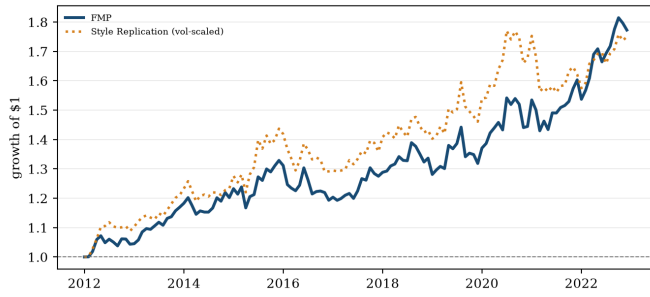


Figure: Common-window cumulative growth of \$1.

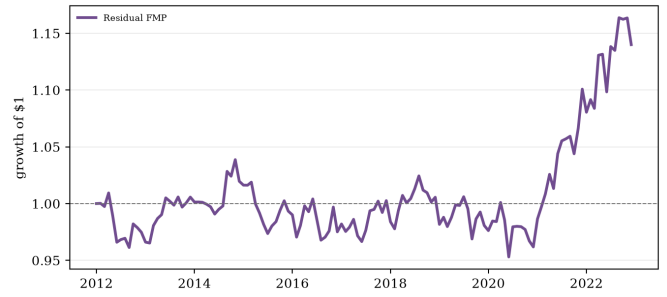


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
momentum_21_252	5.60%	8.38%	0.67	2.87	-10.23%	0.678	1.09%	8.38%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
momentum_21_252	0.02 (0.77)	-0.06 (-1.11)	-0.04 (-0.70)	0.53 (9.02)	0.19 (3.21)	-0.09 (-1.65)	1.31% (1.03)	0.679

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
momentum_21_252	0.30%	0.05%	0.09%	2.68%	1.69%	-0.53%	1.31%	5.60%

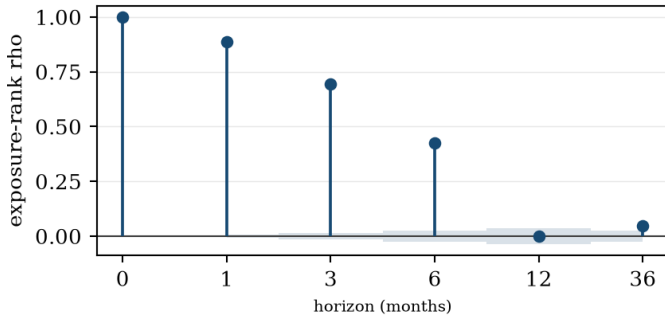


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

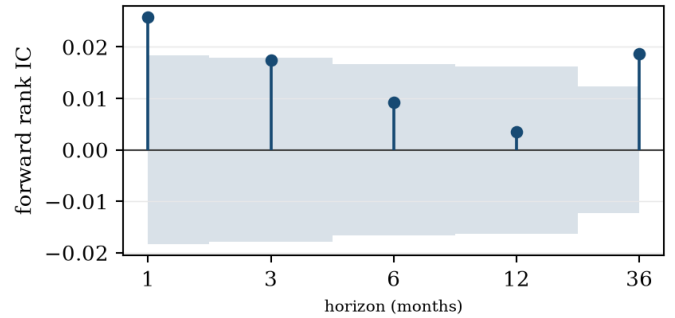


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	momentum_21_252(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.67
Bootstrap Sharpe	0.67 [0.26, 1.13]; Pr(S<=0)=0.0%
Bootstrap tails	skew=-0.15 [-0.54, 0.21]; max DD=-10.2% [-18.1%, -7.3%]

19. share_turnover_21d

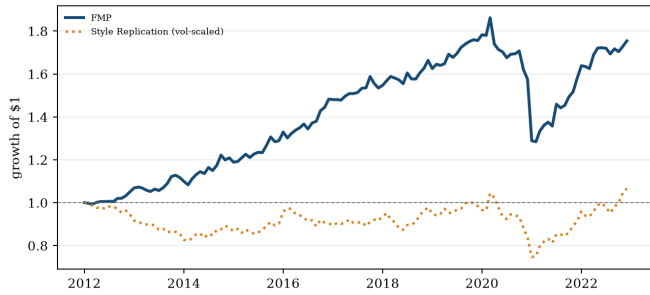


Figure: Common-window cumulative growth of \$1.

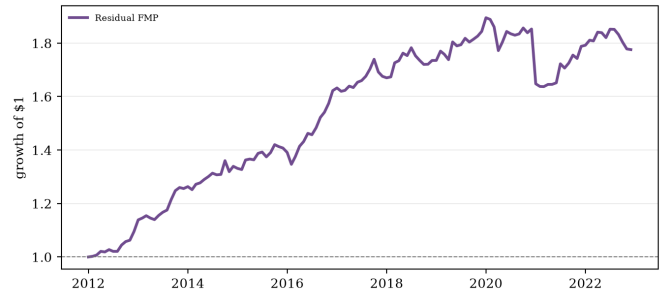


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
share_turnover_21d	5.54%	8.54%	0.65	2.05	-31.05%	0.463	4.89%	8.54%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
share_turnover_21d	-0.01 (-0.23)	-0.31 (-2.63)	-0.00 (-0.05)	-0.03 (-0.54)	-0.27 (-1.48)	0.44 (3.61)	5.47% (2.97)	0.469

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
share_turnover_21d	-0.16%	0.27%	0.01%	-0.17%	-2.39%	2.52%	5.47%	5.54%

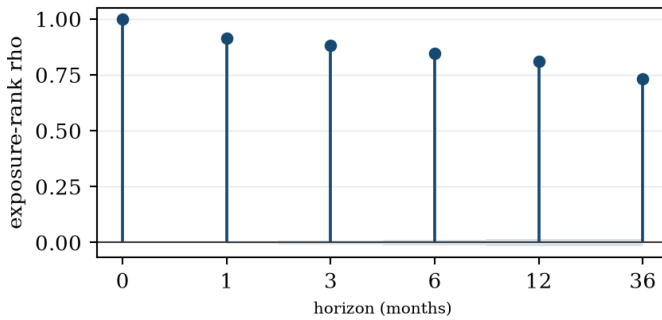


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

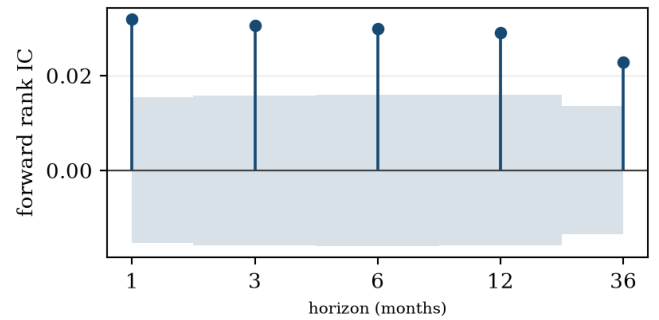


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	share_turnover_21d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.65
Bootstrap Sharpe	0.65 [-0.03, 1.70]; Pr(S<=0)=2.9%
Bootstrap tails	skew=-3.33 [-4.26, 0.68]; max DD=-31.1% [-45.8%, -3.7%]

20. asset_turnover

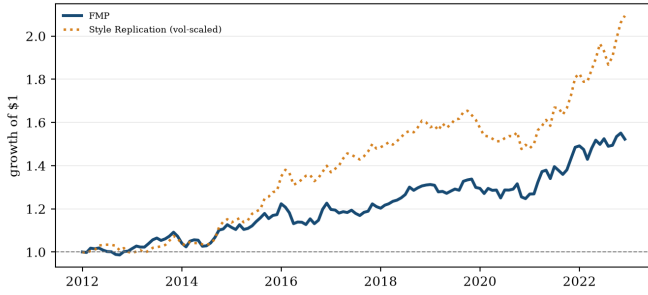


Figure: Common-window cumulative growth of \$1.

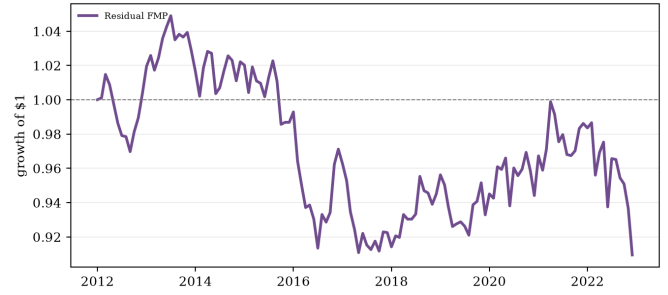


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
asset_turnover	4.05%	6.26%	0.65	2.58	-7.85%	0.487	-0.82%	6.26%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
asset_turnover	0.08 (2.45)	0.17 (3.03)	0.03 (0.75)	-0.01 (-0.20)	0.08 (1.35)	0.59 (10.05)	-0.77% (-0.56)	0.487

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
asset_turnover	1.03%	-0.15%	-0.07%	-0.04%	0.68%	3.37%	-0.77%	4.05%

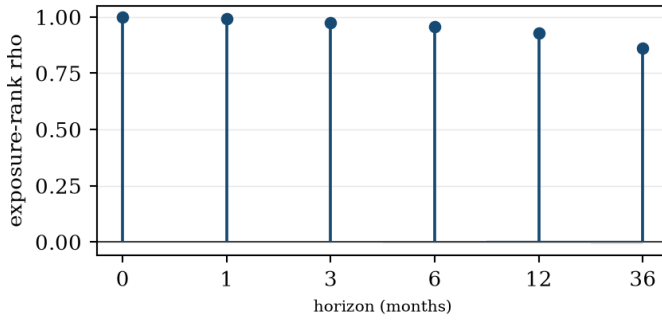


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

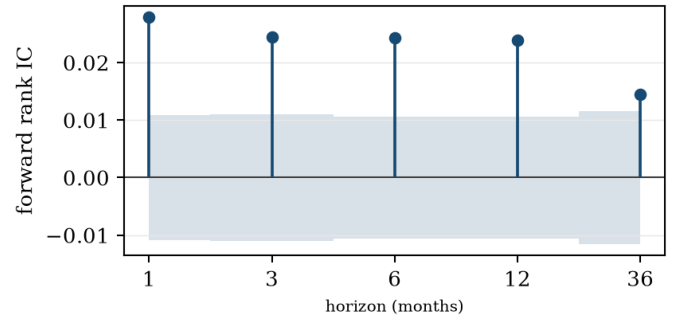


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	asset_turnover(+1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1246 / 1571 / 2083
Available-history Sharpe	0.65
Bootstrap Sharpe	0.65 [0.26, 1.03]; Pr(S<=0)=0.1%
Bootstrap tails	skew=-0.03 [-0.31, 0.21]; max DD=-7.9% [-13.9%, -6.2%]

21. ma_dist_63d

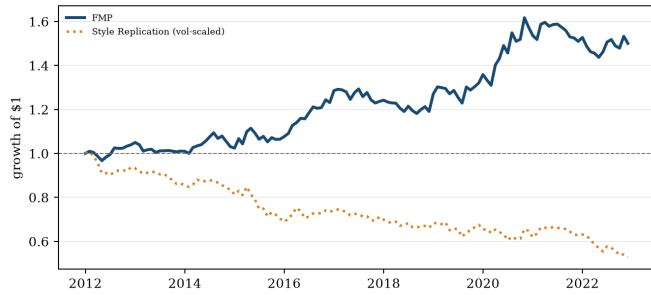


Figure: Common-window cumulative growth of \$1.

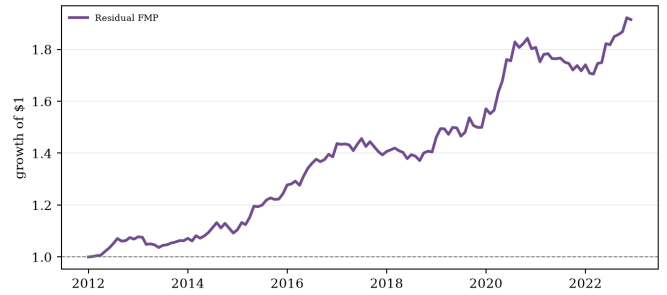


Figure: Cumulative growth after simultaneous control-factor orthogonalisation.

Factor	Ann. ret.	Ann. vol.	Sharpe	HAC t	Max DD	Style R2	Style alpha	Style vol.
ma_dist_63d	4.00%	7.57%	0.53	1.94	-11.14%	0.438	7.66%	7.57%

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	R2
ma_dist_63d	0.13 (2.92)	-0.05 (-0.80)	-0.03 (-0.77)	-0.31 (-6.49)	-0.17 (-3.22)	-0.14 (-2.15)	6.12% (3.74)	0.462

Factor	MKT	SMB	VAL	MOM	BAB	QMJ	Alpha	Total
ma_dist_63d	1.66%	0.05%	0.07%	-1.57%	-1.53%	-0.80%	6.12%	4.00%

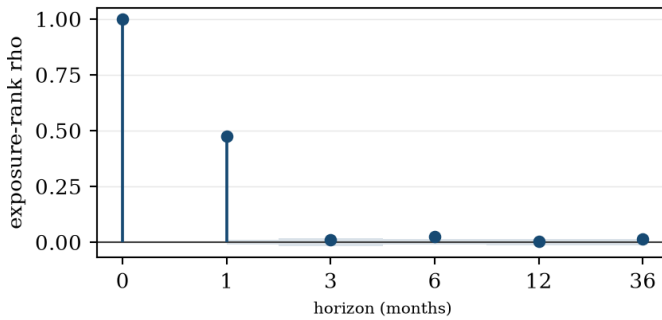


Figure: Exposure-rank ACF; 5% significant only outside the pale HAC band.

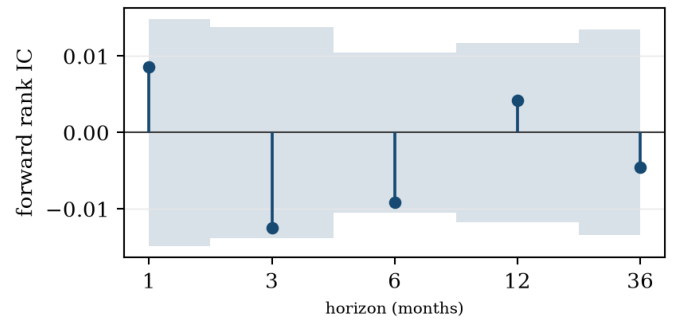


Figure: Exposure-return rank CCF; 5% significant only outside the pale HAC band.

Item	Value
Direction	+1
Common months	131
Signals	ma_dist_63d(-1)
Effective construction	rank/rank; 1%-99%; signal_weighted; LS; beta-neutral=True; group=sector; cost=0bp
Components required	1 of 1
Available / comparison months	131 / 131
Available window	2012-02-01 to 2022-12-01
Dropped: low breadth / common	0 / 0
Stocks, min / median / max	1268 / 1582 / 2103
Available-history Sharpe	0.53
Bootstrap Sharpe	0.53 [0.08, 0.95]; Pr(S<=0)=1.1%
Bootstrap tails	skew=0.92 [0.36, 1.17]; max DD=-11.1% [-20.1%, -6.4%]

Reproducibility

Run ID: 20260906T115339Z_no-clustering-bootstrap-screen-sharpe-0-5-is-2012-2022_036f9a8c5b.

Gold snapshot: version=20260829T051952Z.

Configuration hash: 036f9a8c5b.

Symbols requested / loaded: 0 / 3009.

Seed: 0. Comparison months: 131, 2012-02-01 to 2022-12-01.

Command:

```
C:/Users/User/Desktop/alpha/.venv/Scripts/python.exeC:/Users/User/Desktop/alpha/src/experiments/automatic_factor_grouping/screen_handoff.py--screen-sourcedata/outputs/automatic_factor_grouping/signal_screening/20260906T064949Z_e2fdcf7479--panel-sourcedata/outputs/automatic_factor_grouping/runs/20260906T065013Z_25145edccf--cutoff0.5.
```

Direction is frozen upstream; this report performs no return-based sign selection. The reported direction is applied consistently before every downstream regression, style calculation, plot, table, PCA coordinate, and index assignment. The run directory contains the source configuration, normalized configuration, manifest, canonical tables, monthly returns, PCA coordinates, plots, report source, and PDF.